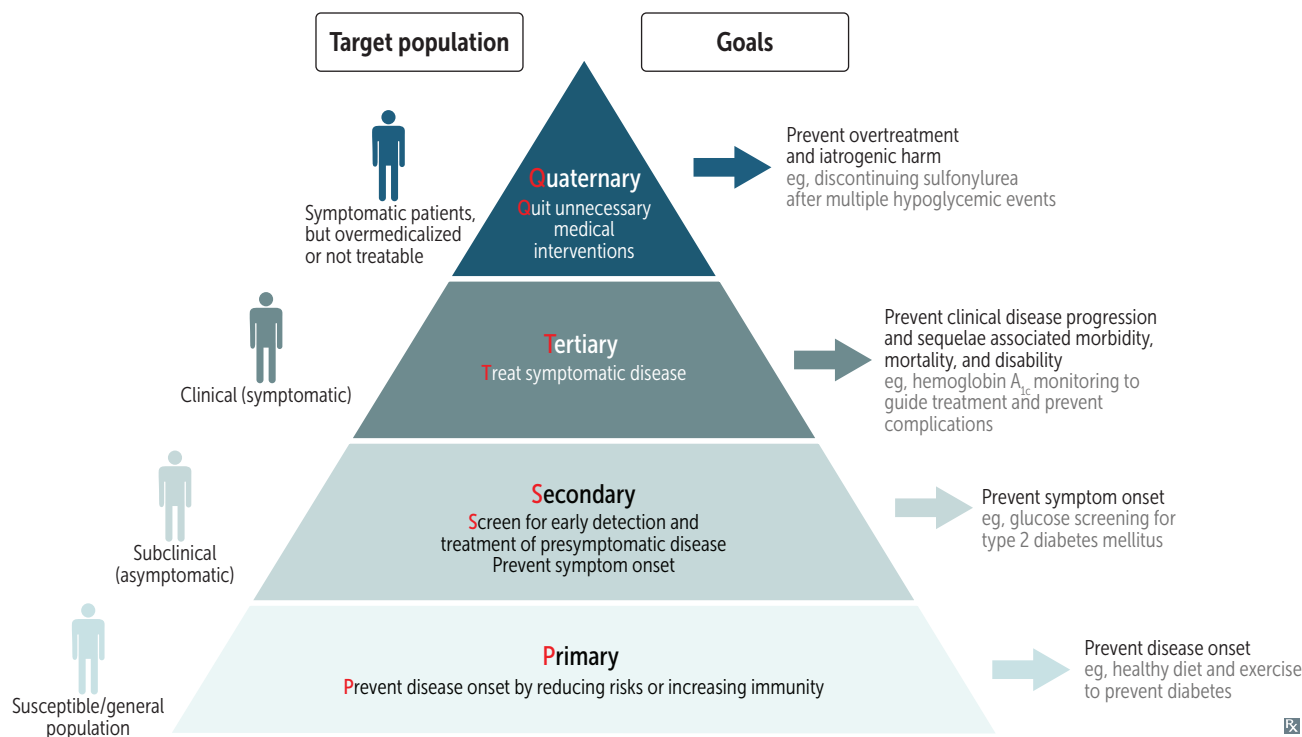


► PUBLIC HEALTH SCIENCES—HEALTHCARE DELIVERY

Disease prevention



Major medical insurance plans

Exclusive provider organization (EPO)	Requires in-network care; out-of-network coverage available for emergencies.
Health maintenance organization (HMO)	Lower premiums, limited to in-network providers, requires primary care referral for specialists, lower out-of-pocket cost.
Point of service (PoS)	Combination of HMO and PPO, allows out-of-network care at higher cost, and requires primary care referrals.
Preferred provider organization (PPO)	Flexibility with in-network and out-of-network care, higher cost for out-of-network services, no referral requirement.
Accountable care organization (ACO)	Group of providers who voluntarily coordinate care for Medicare patients.
High deductible health plan (HDHP)	Low premiums, high out-of-pocket cost, compatible with health savings account (HSA).

Healthcare payment models

Bundled payment	Healthcare organization receives a set amount per service, regardless of ultimate cost, to be divided among all providers and facilities involved.
Capitation	Physicians receive a set amount per patient assigned to them per period of time, regardless of how much the patient uses the healthcare system. Used by some HMOs.
Discounted fee-for-service	Insurer and/or patient pays for each individual service at a discounted rate predetermined by providers and payers (eg, PPOs).
Fee-for-service	Insurer and/or patient pays for each individual service.
Global payment	Insurer and/or patient pays for all expenses associated with a single incident of care with a single payment. Most commonly used during elective surgeries, as it covers the cost of surgery as well as the necessary pre- and postoperative visits.

Medicare and Medicaid

Medicare and Medicaid—federal social healthcare programs that originated from amendments to the Social Security Act. Medicare is available to patients ≥ 65 years old, < 65 with certain disabilities, and those with end-stage renal disease. Medicaid is joint federal and state health assistance for people with limited income and/or resources.

Medicar**E** is for **E**lderly.
Medicai**D** is for **D**isadvantaged.

The 4 parts of Medicare:

- Part **A**: hospital **A**dmissions, including hospice, skilled nursing
- Part **B**: **B**asic medical **b**ills (eg, physician fees, diagnostic testing)
- Part **C**: (parts A + B = **C**ombo) delivered by approved private **c**ompanies
- Part **D**: prescription **D**rugs

Palliative care

Medical care aiming to provide comfort, relieve suffering, and improve quality of life in patients with complex or life-threatening illness regardless of their diagnosis or prognosis. Often concurrent with curative or life-prolonging treatment.

Delivered by interdisciplinary team (eg, physicians, nurses, social workers) in hospitals, outpatient clinics, or at home.

Hospice care (end-of-life care)—form of palliative care for patients with prognosis ≤ 6 months when curative or life-prolonging treatment is no longer wanted or beneficial.

Types of medical errors

	May involve patient identification, diagnosis, monitoring, healthcare-associated infection, medications, procedures, devices, documentation, handoffs. Medical errors should be disclosed to patients, independent of immediate outcome (harmful or not).	
Active error	Occurs at level of frontline operator (eg, wrong IV pump dose programmed).	Immediate impact.
Latent error	Occurs in processes indirect from operator but impacts patient care (eg, different types of IV pumps used within same hospital).	Accident waiting to happen.
Never event	Adverse event that is identifiable, serious, and usually preventable (eg, scalpel retained in a surgical patient's abdomen).	Major error that should never occur. Sentinel event —a never event that leads to death, permanent harm, or severe temporary harm.
Near miss	Unplanned event that does not result in harm but has the potential to do so (eg, pharmacist recognizes a medication interaction and cancels the order).	Narrow prevention of harm that exposes dangers.

Medical error analysis

	DESIGN	METHODS
Root cause analysis	R etrospective approach. Applied after failure event to prevent recurrence.	Uses records and participant interviews (eg, 5 whys approach, fishbone/cause-and-effect diagrams, process maps) to identify all the underlying problems (eg, process, people, environment, equipment, materials, management) that led to an error.
Failure mode and effects analysis	F orward-looking approach. Applied before process implementation to prevent failure occurrence.	Uses inductive reasoning to identify all the ways a process might fail and prioritizes them by their probability of occurrence and impact on patients.

Causes of medical errors

Burnout	Prolonged, excessive stress leading to emotional exhaustion, depersonalization, leading to reduced professional efficacy.
Fatigue	Sleep/rest deprivation resulting in cognitive impairment and decreased attention to detail.